Reunited with his wife and son after Partition, Talwar came to New Delhi. After only a week, he was posted to Kanpur. After eleven days at Kanpur, he was shifted to Jamshedpur. This too proved to be only a temporary posting and within three weeks he was posted to the branch department in the Local Head Office at Calcutta in early 1948.[2] Talwar narrates in his recollections how the Local Head Office (LHO) would ask the district manager in Lahore to transfer an assistant or two from Lahore to Calcutta and how he would dread being picked up for such a transfer. Unhappy at the constant transfers, Talwar met Puri, then the Secretary and Treasurer at Calcutta, and requested that he be allowed to stay at one place for at least six months. Puri told him that his request would only be heard once he carried out the transfer to Calcutta since in the 'Imperial Bank orders are carried out, not talked about.'[3] Talwar did so, and this move ushered in a period of long stability with Talwar staying in Calcutta for more than twelve years.

Talwar came to the branch department as a Junior Assistant. He worked there for many years learning the work of each of its sections. He was promoted to Assistant Superintendent and after a couple of years made Superintendent. Talwar worked in the Loans Department for more than three years. His boss, a Superintendent called Gupta, was a hard taskmaster with an abrasive personality to boot. Talwar worked very hard to master the various nuances of banking especially in the area of credit. It is striking that Talwar, who eventually became the best Chairman the SBI has had, never worked in a branch. However, his reorganization exercise showed that he understood the problems of a branch manager better than most of his peers, including those who had occupied such a position. Talwar recalled that however hard he tried, he could never measure up to Gupta's expectations. But this experience contributed to Talwar making every effort to excel in his work and his subsequent rapid rise in the bank's hierarchy. Talwar acknowledged this when he observed: 'Talwar was made in Calcutta by Gupta.'[4]

Talwar became Superintendent of Branches and Superintendent of Advances in the Bengal Circle of SBI and was also Inspector of Branches under the Central Office. In 1961, he was appointed Deputy Secretary and Treasurer of the Bengal Circle. A year later, he moved to the Madras Circle